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  ],
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  "framework" : "Swift",
  "identifier" : "/documentation/Swift/CustomStringConvertible",
  "metadataVersion" : "0.1.0",
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      "Swift"
    ],
    "preciseIdentifier" : "s:s23CustomStringConvertibleP"
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  "title" : "CustomStringConvertible"
}
-->

CustomStringConvertible

A type with a customized textual representation.

```
protocol CustomStringConvertible
```

Overview

Types that conform to the `CustomStringConvertible` protocol can provide their own representation to be used when converting an instance to a string. The `String(describing:)` initializer is the preferred way to convert an instance of *any* type to a string. If the passed instance conforms to `CustomStringConvertible`, the `String(describing:)` initializer and the `print(_:)` function use the instance’s custom `description` property.

Accessing a type’s `description` property directly or using `CustomStringConvertible` as a generic constraint is discouraged.

Conforming to the CustomStringConvertible Protocol

Add `CustomStringConvertible` conformance to your custom types by defining a `description` property.

For example, this custom `Point` struct uses the default representation supplied by the standard library:

```
struct Point {
    let x: Int, y: Int
}

let p = Point(x: 21, y: 30)
```

print(p)
// Prints "Point(x: 21, y: 30)"
```

After implementing the `description` property and declaring `CustomStringConvertible` conformance, the `Point` type provides its own custom representation.

```
extension Point: CustomStringConvertible {
    var description: String {
        return "(\(x), \(y))"
    }
}

print(p)
// Prints "(21, 30)"
```

Relationships

Conforming Types

[`SIMD16`](/documentation/Swift/SIMD16)

[`Mirror`](/documentation/Swift/Mirror)

[`Bool`](/documentation/Swift/Bool)

[`KeyValuePairs`](/documentation/Swift/KeyValuePairs)

[`Substring`](/documentation/Swift/Substring)

[`Values-swift.struct`](/documentation/Swift/Dictionary/Values-swift.struct)

[`DefaultStringInterpolation`](/documentation/Swift/DefaultStringInterpolation)

[`Float16`](/documentation/Swift/Float16)

[`ArraySlice`](/documentation/Swift/ArraySlice)

[`UTF8View`](/documentation/Swift/String/UTF8View)

[`DiscontiguousSlice`](/documentation/Swift/DiscontiguousSlice)

[`SIMDMask`](/documentation/Swift/SIMDMask)

[`UnicodeScalarView`](/documentation/Swift/String/UnicodeScalarView)

[`ContiguousArray`](/documentation/Swift/ContiguousArray)

[`Duration`](/documentation/Swift/Duration)

[`Character`](/documentation/Swift/Character)

[`Int64`](/documentation/Swift/Int64)

[`Kind-swift.struct`](/documentation/Swift/Unicode/UTF8/ValidationError/Kind-swift.struct)

[`Float`](/documentation/Swift/Float)
```

[`UInt64`](/documentation/Swift/UInt64)

[`AtomicStoreOrdering`](/documentation/Synchronization/AtomicStoreOrdering)

[`Float80`](/documentation/Swift/Float80)

[`SIMD64`](/documentation/Swift/SIMD64)

[`UInt16`](/documentation/Swift/UInt16)

[`Scalar`](/documentation/Swift/Unicode/Scalar)

[`SIMD2`](/documentation/Swift/SIMD2)

[`Range`](/documentation/Swift/Range)

[`UInt`](/documentation/Swift/UInt)

[`Array`](/documentation/Swift/Array)

[`Int16`](/documentation/Swift/Int16)

[`SIMD32`](/documentation/Swift/SIMD32)

[`SIMD4`](/documentation/Swift/SIMD4)

[`SIMD3`](/documentation/Swift/SIMD3)

[`Ranges-swift.struct`](/documentation/Swift/RangeSet/Ranges-swift.struct)

[`Encoding`](/documentation/Swift/String/Encoding)

[`RemoteCallTarget`](/documentation/Distributed/RemoteCallTarget)

[`Dictionary`](/documentation/Swift/Dictionary)

[`RangeSet`](/documentation/Swift/RangeSet)

[`Set`](/documentation/Swift/Set)

[`TaskLocal`](/documentation/Swift/TaskLocal)

[`Int128`](/documentation/Swift/Int128)

[`WordPair`](/documentation/Synchronization/WordPair)

[`SIMD8`](/documentation/Swift/SIMD8)

[`AnyHashable`](/documentation/Swift/AnyHashable)

[`UTF16View`](/documentation/Swift/String/UTF16View)

[`String`](/documentation/Swift/String)

[`Index`](/documentation/Swift/DiscontiguousSlice/Index)

[`UInt128`](/documentation/Swift/UInt128)

[`Double`](/documentation/Swift/Double)

[`StaticString`](/documentation/Swift/StaticString)

[`AtomicUpdateOrdering`](/documentation/Synchronization/AtomicUpdateOrdering)

[`UnownedJob`](/documentation/Swift/UnownedJob)

[`Int8`](/documentation/Swift/Int8)

[`Int32`](/documentation/Swift/Int32)

[`ClosedRange`](/documentation/Swift/ClosedRange)

[`ValidationError`](/documentation/Swift/Unicode/UTF8/ValidationError)

[`Keys-swift.struct`](/documentation/Swift/Dictionary/Keys-swift.struct)

[`Int`](/documentation/Swift/Int)

[`AtomicLoadOrdering`](/documentation/Synchronization/AtomicLoadOrdering)

[`UInt8`](/documentation/Swift/UInt8)

[`UInt32`](/documentation/Swift/UInt32)

[`TaskPriority`](/documentation/Swift/TaskPriority)

Inherited By

[`FixedWidthInteger`](/documentation/Swift/FixedWidthInteger)

[`SIMD`](/documentation/Swift/SIMD)

[`BinaryInteger`](/documentation/Swift/BinaryInteger)

[`UnsignedInteger`](/documentation/Swift/UnsignedInteger)

[`SignedInteger`](/documentation/Swift/SignedInteger)

[`LosslessStringConvertible`](/documentation/Swift/LosslessStringConvertible)

[`StringProtocol`](/documentation/Swift/StringProtocol)

[`CodingKey`](/documentation/Swift/CodingKey)
